Continuous Functions

Bartle & Sherbert §5.1, §5.3 (5.3.1–5.3.5, 5.3.7–5.3.10), §5.4 (5.4.1–5.4.14) · FYUGP pattern (April 2026), doubled

HOW THIS PAPER WORKS

A real FYUGP paper has 10 short (3 marks), 8 paragraph (6 marks) and 2 essay (10 marks, answer one) questions. This paper has **twice as many** (20 + 16 + 4), all from Module 1. Each of the module's 40 concepts is the main point of exactly one question.

Time	4 hours (or two sittings of 2 hours)
Maximum	140 marks (A ceiling 48, B ceiling 72, C answer any two)

PYQ MTS-25 Q9 = MTS 6B 10, March 2025, question 9 (old CBCSS course, same textbook). "also" = another paper asked the same thing. "adapted" = a 1-mark MCQ turned into a 3-mark question. **CALC-24** = related FYUGP Calculus paper (Nov 2024).

Section A · Short answer

All questions can be answered. Each carries 3 marks. Ceiling 48.

1. Define continuity of a function $f : A \to \mathbb{R}$ at a point $c \in A$. Show that a constant function is continuous on $\mathbb{R}$.

PYQ MTS-22 Q1

2. State the neighbourhood form of continuity, and the limit condition when c is a cluster point of A. Why is every function continuous at an isolated point of its domain?

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3. State the Discontinuity Criterion. Use it to show that $f(x) = [x]$ is discontinuous at $x = 1$.

PYQ MTS-26 Q1

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4. Show that $f(x) = |x|$ is continuous at every point of $\mathbb{R}$.

CALC-24 Q12 · related FYUGP

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5. Define continuity on a set $B \subseteq A$. Prove from the ε–δ definition that $g(x) = x$ is continuous on $\mathbb{R}$.

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6. Is the function $f(x) = x^2$ for $x \leq 2$, $f(x) = x + 10$ for $x > 2$, continuous at 2? Why? What change would make it continuous?

CALC-24 Q3 · related FYUGP

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7. (a) Is $1/x$ continuous at 0? (b) Show sgn is not continuous at 0 and name the kind of break. (c) Which of $\sin x$, $\sin(1/x)$, $1/x$ are continuous on $A = \{x \geq 0\}$?

PYQ MAT-22 MCQ 3, 8 · adapted **also MAT-21 MCQ 6**

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8. Can $f(x) = \dfrac{x^2 + x - 6}{x - 2}$ $(x \neq 2)$ be defined at 2 so that it is continuous there? Show that $F(x) = x \sin(1/x)$ $(x \neq 0)$, $F(0) = 0$, is continuous at 0.

PYQ MAT-22 MCQ 9 · adapted

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9. Define a bounded function. Give an example of a continuous function that is not bounded, and prove it.

PYQ MTS-26 Q2 also MAT-22 MCQ 6

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10. Show by examples that the Boundedness Theorem fails if (i) the interval is unbounded, (ii) the interval is not closed, (iii) f is not continuous. Give a continuous function on a non-closed interval with no absolute maximum.

PYQ MTS-22 Q2b also MAT-21 Q13b, Q14b

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11. Define absolute maximum and absolute minimum of f on A. Does $1/x$ have either on $(0, \infty)$? Find them for x^2 on [−1, 1].

PYQ MAT-22 MCQ 10 · adapted

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12. Using the Location of Roots Theorem, show that (a) $xe^x - 2 = 0$ has a root in (0, 1); (b) $6x^3 - 3x^2 + 2$ has a zero in (−1, 1).

PYQ MAT-21 Q17b

also MAT-22 MCQ 12, MAT-21 MCQ 7

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13. State Bolzano's Intermediate Value Theorem. Use it to show that some $c \in (1, 2)$ has $c^2 = 2$.

PYQ MAT-21 Q1

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14. Define uniform continuity. Is every continuous function uniformly continuous? Is every uniformly continuous function continuous?

PYQ MAT-22 Q1

also MAT-21 Q2, MAT-22 MCQ 13, MAT-21 MCQ 8

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15. Define a Lipschitz function and explain it geometrically. Show that $f(x) = 3x + 1$ is Lipschitz.

PYQ MTS-26 Q3 also MAT-21 Q28a

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16. Give an example of a uniformly continuous function that is not Lipschitz, and justify.

PYQ MTS-25 Q2 also MAT-22 Q15

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17. If f and g are Lipschitz on A, show that $f + g$ is Lipschitz. Give an example of a Lipschitz f for which f^2 is not Lipschitz.

PYQ MTS-22 MCQ 3 · adapted

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18. Show that $f(x) = \dfrac{1}{1 + x^2}$ is uniformly continuous on $\mathbb{R}$.

PYQ MTS-22 MCQ 4 · adapted

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19. Define a step function. Is s = 0 on [−2, −1), 1 on [−1, 0], ½ on (0, ½), 3 on [½, 1), −2 on [1, 3], 2 on (3, 4] a step function on [−2, 4]? Is $f(x) = x$ on [0, 1] a step function?

PYQ MAT-22 Q5 · also MAT-21 Q3

20. State the Weierstrass Approximation Theorem. Why can it not hold for $1/x$ on (0, 1]?

PYQ MTS-25 Q3 · also MAT-22 Q2

Section B · Paragraph / problem

All questions can be answered. Each carries 6 marks. Ceiling 72.

21. State and prove the Sequential Criterion for continuity.

22. Prove from the ε–δ definition that x^2 is continuous at every $c \in \mathbb{R}$, giving δ in terms of ε and c. Hence prove that polynomial functions are continuous on $\mathbb{R}$.

PYQ MTS-25 Q1

23. Prove that $1/x$ is continuous on $(0, \infty)$ using ε–δ. Find an admissible δ for $c = \frac{1}{2}$, $\varepsilon = \frac{1}{10}$.

24. Prove that Dirichlet's function is discontinuous at every point of $\mathbb{R}$.

PYQ MTS-25 Q16

25. Define Thomae's function on $(0, \infty)$ and prove that it is continuous at every irrational and discontinuous at every rational point.

PYQ MTS-22 MCQ 1 · adapted

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26. State and prove the Boundedness Theorem.

PYQ MTS-26 Q16

also MAT-22 Q27, MAT-21 Q13a, MTS-22 Q2a

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27. Let $f : [a, b] \to \mathbb{R}$ be continuous and $\inf f(I) \le k \le \sup f(I)$. Prove that $f(c) = k$ for some $c \in I$. Illustrate with $f(x) = x^3 - 3x$ on [−1.5, 1.5] and k = 1.

PYQ MAT-22 Q13

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28. Prove that the image of a closed bounded interval under a continuous function is a closed bounded interval. Show by an example that it need not be $[f(a), f(b)]$.

PYQ MAT-21 Q15

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29. State and prove the Preservation of Intervals Theorem. Find $f((-1, 1))$ and $f([0, \infty))$ for $f(x) = \dfrac{1}{x^2 + 1}$.

PYQ MTS-25 Q17 **also MAT-22 Q14**

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30. State the Nonuniform Continuity Criteria. Show that $1/x$ is not uniformly continuous on $(0, \infty)$. Is it uniformly continuous on $[a, 1)$, $a > 0$?

PYQ MAT-21 Q16 **also MTS-22 MCQ 2**

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31. Prove that a Lipschitz function on A is uniformly continuous on A. Find δ for $\sin x$ on $\mathbb{R}$.

PYQ MAT-21 Q28b

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32. Prove that $g(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$ (in particular on [0, 2]).

PYQ MTS-26 Q24b **also MAT-22 Q36b**

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33. Show that if f and g are uniformly continuous on $\mathbb{R}$, then $f + g$ is, but the product fg need not be.

PYQ MAT-21 Q22

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34. If f is uniformly continuous on A and (x_n) is a Cauchy sequence in A, prove that $(f(x_n))$ is Cauchy. Show this can fail for a function that is merely continuous.

PYQ MAT-21 Q36a **also MTS-22 Q3**

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35. Let f be continuous on [a, b] and ε > 0. Prove that there is a step function s_ε with $|f(x) - s_\varepsilon(x)| < \varepsilon$ on [a, b]. Find m and s_ε for $f(x) = x$ on [0, 1], ε = 0.1.

PYQ MTS-26 Q17

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36. Define a piecewise linear function. Prove that a continuous f on [a, b] can be approximated within ε by a continuous piecewise linear function. For x^2 on [0, 1] with nodes 0, ¼, ½, ¾, 1, find the largest error.

Section C · Essay

Answer any two. Each carries 10 marks.

37. State and prove the Location of Roots Theorem.

PYQ MTS-25 Q24 also MAT-21 Q17a

38. State and prove the Maximum–Minimum Theorem.

PYQ MAT-22 Q36a

also MAT-21 Q14a, MAT-22 MCQ 11

39. State and prove the Uniform Continuity Theorem.

PYQ MTS-26 Q24a also MAT-22 Q28, MAT-21 Q27

40. State and prove the Continuous Extension Theorem.

PYQ MAT-22 Q29

Coverage

TOPIC	QUESTIONS
T1 ε–δ definition	1, 5
T2 Neighbourhoods, limits	2
T3 Sequential criterion	3, 21
T4 ε–δ proofs	4, 22, 23
T5 Where functions break	6, 7
T6 Dirichlet, Thomae	24, 25
T7 Filling holes	8
T8 Boundedness	9, 10, 26
T9 Max–Min	10, 11, 38
T10 Location of Roots	12, 37
T11 Bolzano	13, 27
T12 Images of intervals	28, 29
T13 Uniform continuity	14
T14 Not uniformly continuous	30
T15 UC Theorem	39
T16 Lipschitz	15, 16, 17, 18, 31
T17 √x	32
T18 Sums, products	33
T19 Cauchy sequences	34
T20 Continuous extension	40
T21 Step, piecewise linear	19, 35, 36
T22 Weierstrass	20

All 55 in-syllabus past questions appear (40 written, 15 MCQs), plus 2 related FYUGP ones.